

v5 integer-sample timing-offset report

At the frozen **one-sample** timing offsets ($\pm 1/64$ symbol, or $\pm 1.5625\%$), the existing nominal decoder passed **3,256/12,000 (27.13%; 95% Wilson interval 26.35–27.94%)** for an early window and **3,231/12,000 (26.92%; 26.14–27.73%)** for a late window, with IID Gaussian sample-noise SD 0.45 . At **± 2 samples** ($\pm 3.125\%$ symbol) and all larger tested magnitudes, it passed **0/12,000** frames in each direction; each zero-success 95% Wilson interval has an upper bound near **0.032%**. At zero offset it passed **11,382/12,000 (94.85%; 94.44–95.23%)**.

This was meaningful as a **receiver-window test under the project's discrete 64-sample framing**. The impairment is a constant integer sample offset: the existing receiver takes each 64-sample window early or late, at the same offset for all 32 symbols. The serialized frame has 64 zero-valued guard samples on both edges, so misaligned first/last windows are defined without circular wrapping. The waveform, signed encoder, carrier, sample rate, symbol framing, and matched-filter implementation are unchanged. There is no CFO, sample-rate drift, interpolation, or pulse shaping. The abrupt segment boundaries and guards are explicit software assumptions, not physical channel features.

The timing-aware **oracle** knew the exact offset and reselected the true symbol windows before using the nominal projection. It passed **94.55–95.05%** across the noisy offset grid. The analytic prediction was 94.78% at each offset; every prediction fell inside its observed pooled Wilson interval, and the largest empirical/theory gap was **0.27 percentage points**. For clean frames, the existing decoder passed **6/6** fixed vectors at offsets $-1, 0, +1$ and **0/6** at ± 2 and beyond; the timing-aware oracle passed **6/6** at every tested offset. The always-zero weak sanity floor passed **0/6** fixed vectors and is not a competing decoder.

The fixed run used the six v2 input vectors, timing offsets $-8, -4, -2, -1, 0, +1, +2, +4, +8$ samples, clean and $\sigma=0.45$ controls, 2,000 trials per vector/offset, and seed 20261007 : **108,000 noisy input frames**. Each noisy frame was shared by the nominal and oracle methods. At ± 1 sample, per-vector pass shares ranged from **11.60% to 62.75%** for early windows and **10.70% to 62.25%** for late windows, so the pooled results vary materially across the six selected patterns. Wilson intervals quantify only random-noise variation conditional on this fixed panel. They do not capture model or physical uncertainty.

The experiment follows established communications practice in treating symbol timing as a receiver synchronization problem: MathWorks documents `comm.SymbolSynchronizer` as correcting timing clock skew and demonstrates a fixed timing error ([🔥 MathWorks documentation](#)); a Stanford communications text has a dedicated symbol-timing recovery section ([🌐 Fundamentals of Synchronization, § 6.3](#)). A separate MathWorks link example distinguishes sample-clock offset from carrier-frequency offset and describes timing

recovery ( [CCSDS optical-link example](#)). These sources justify testing timing separately; they do **not** make this discrete offset grid a physical channel model.

The v5 conditions were fixed before the full run in the [protocol lock](#). This is an **arbitrary discrete sample-offset software stress test**, not a measured channel or evidence of physical validity. All **47 project tests passed**, the JSON manifest validates, and pre-run SHA-256 checks confirm the v1–v4 sources, tests, reports, and result artifacts remain unchanged.

- [v5 runner](#)
- [v5 tests](#)
- [Pre-run protocol lock](#)
- [JSON results and protocol](#)
- [CSV results](#)
- [Summary plot](#)
- [Pre-run protected-file checksums](#)
- [v5 output checksums](#)